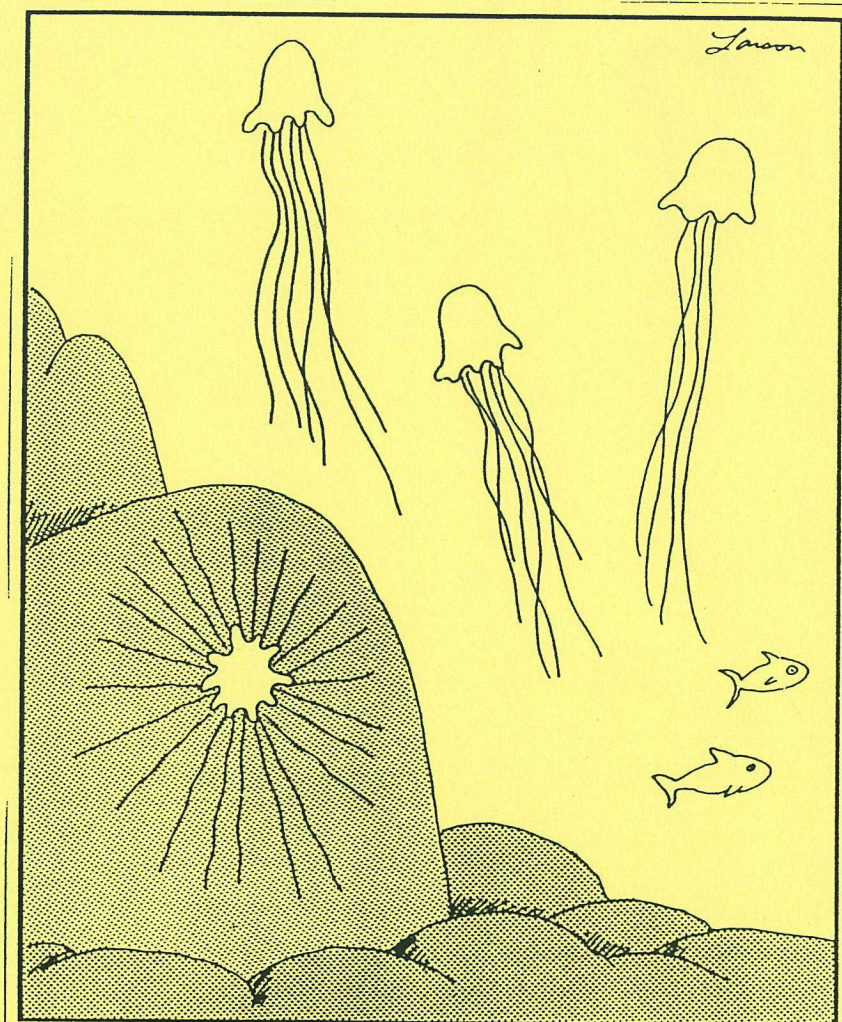


Year 10 Science

PHYSICS I



When jellyfish travel at unsafe speeds

Motion

Section A MCQ 15 marks

Section B Short Answers..... 30 marks

Total 45 marks

SECTION A: Multiple Choice

$$g = 10 \text{ ms}^{-2}$$

1. A vector has

- (a) direction only
- (b) magnitude only
- (c) both direction and magnitude
- (d) neither magnitude or direction
- (e) none of the above

2. Which one of these is a vector quantity?

- (a) area
- (b) speed
- (c) energy
- (d) force
- (e) none of the above

3. The velocity of a car, coasting on a level road after the motor has been shut off is

- (a) zero
- (b) constant but not zero
- (c) increasing at a constant rate
- (d) decreasing at a constant rate
- (e) none of the above

4. The average velocity of a car if it travels a distance of 270 km North in 6 hours is

- (a) 8 ms^{-1}
- (b) 35 kmhr^{-1}
- (c) 0.087 ms^{-1}
- (d) 12.5 ms^{-1}
- (e) none of the above

5. During an experiment a student using a 50 Hz tic-a-timer measured the distance between the 11th and 19th dots on a tape to be 14 cm. The average velocity for this section was

- (a) 8.75 cms^{-1}
- (b) 8.75 ms^{-1}
- (c) 0.087 ms^{-1}
- (d) 87.5 cms^{-1}
- (e) none of the above

6. A car travelling at 16 ms^{-1} accelerates at 3.3 ms^{-2} for 7 s. Which of the following is true.

- (a) the initial velocity is not known
- (b) 16 ms^{-1} is the initial velocity the final velocity can't be found
- (c) the final velocity is 45.7 ms^{-1}
- (d) the final velocity is 143.1 kmhr^{-1}
- (e) none of the above

7. A car travelling at 13 ms^{-1} accelerates at 3 ms^{-2} for 7s. Which of the following is true.

- (a) the initial velocity is not known
- (b) 13 ms^{-1} is the initial velocity and final velocity can't be found
- (c) the final velocity is 34 ms^{-1}
- (d) the car will be going slower after 7s.
- (e) none of the above

8. Which of the following is a SCALAR quantity?

- (a) force
- (b) displacement
- (c) none of these
- (d) velocity
- (e) speed

9. When a "sky diver" free falls he reaches a terminal velocity because

- (a) acceleration due to gravity changes
- (b) wind resistance affects his fall
- (c) his mass determines his velocity
- (d) he does not accelerate
- (e) the statement is false

10. A runner A can run 8 ms^{-1} and runner B can run at 7.5 ms^{-1} . In a 200 m race runner B starts 20 m in front of A. Which of the following is true

- (a) runner B wins by 2 s
- (b) runner A wins by 2 s
- (c) runner B wins by 1 s
- (d) runner a wins by 1 s
- (e) none of the above

11. A body moving at 20 ms^{-1} has its velocity changed to 4 ms^{-1} in 2s. The body's acceleration is

- (a) 8 ms^{-2}
- (b) 12 ms^{-2}
- (c) 08 ms^{-2}
- (d) 20 ms^{-1}
- (e) none of the above

12. A driver noticed that his speedo reading was 118.8 kmhr^{-1} . Which of the following represents the speedo reading in ms^{-1} .

- (a) 11880 ms^{-1}
- (b) 33 ms^{-1}
- (c) 118 ms^{-1}
- (d) 330 ms^{-1}
- (e) none of the above

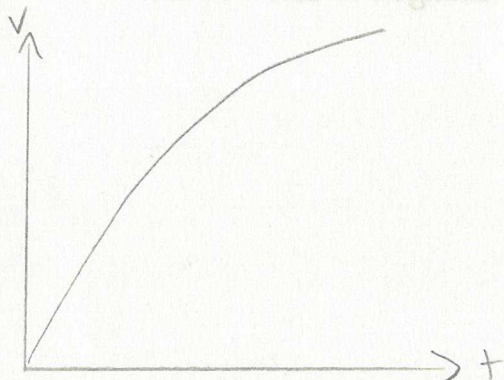
13. The following forces all act on the same point, 13 N West, 15 N East and 8 N West. Their resultant would be given by which single force.

- (a) 6 N West
- (b) 15 N East
- (c) 23 N West
- (d) 36 N West
- (e) 6 N East

14. A force of 160 N acts on a mass of 120 g. Which of the following is the value for acceleration

- (a) 1.33 ms^{-2}
 (b) 19.2 ms^{-2}
 (c) 1333.3 ms^{-2}
 (d) 133.3 ms^{-2}
 (e) none of the above

15.



For the above graph which statement is true

- (a) the velocity is constant
 (b) the acceleration increases
 (c) the displacement is changing at constant rate
 (d) the acceleration is zero
 (e) none of the above

End of Section A

Formula which may be used in Section B

$$V_{\text{avg}} = \frac{u + v}{2}$$

$$a = \frac{v - u}{t}$$

$$V_{\text{avg}} = \frac{\text{total } S}{\text{total } t}$$

$$v = u + at$$

$$S = ut + \frac{1}{2} at^2$$

$$V_{av.} = \frac{u+v}{2}$$

$$V_{av.} = \frac{S(t_{total})}{t(")}$$

$$a = \frac{v-u}{t}$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

Section B - Written Section

1. (a) What are the standard units for length, mass and time in Physics?
- (b) What is the difference between a derived and a fundamental unit?
- (c) List 3 derived units.

2. (a) Place the following in standard scientific notation.

(i) 2732

(ii) 0.00144

(iii) 6254.2

(iv) 7462×10^4

- (b) How many Joules are there in 8 MJ

8 MJ = 8 Mega Joules

- (c) Arrange the following in order of largest to smallest

(a) 76 cm (b) 0.85 cm (c) 7200 mm (d) 0.001 Km

7200 cm 100 cm

3. (a) List 3 non-contact forces.

- (b) Is weight a vector or a scalar? Explain.
If the weight of a body is 770 N what is its mass?

- (c) For the following situations explain whether friction is an advantage or disadvantage.

(i) Walking

(ii) Sliding furniture

(iii) Writing

4. An aeroplane flies east at 250 km hr^{-1} for 3 hours and then south at 200 km hr^{-1} for 1 hour.

WEST

- (a) What is the total distance travelled?
- (b) What is the average speed?
- (c) What is the displacement?
- (d) What is the average velocity?

- ✓ 5. A train is moving with a velocity of 25 ms^{-1} . In a time of 8 s the train slows down to 5 ms^{-1} . What is the deceleration of the train?
6. A person standing on a cliff wants to determine how high it is above the river below. She times how long it takes for a stone to reach the water and finds it take 3.60 s ($g = 10 \text{ ms}^{-2}$).
- ✓ (a) Calculate the velocity with which the stone hits the water.
(b) Determine the height of the cliff.
7. (a) State Newton Second law of Motion.
(b) What formula represents this law.
(c) A ball of mass 0.4 kg is hit by a bat. If the acceleration of the ball is 20 ms^{-2} then find the force of the bat on the ball.
(d) Using the terms action and reaction forces, explain
(a) the recoil of a gun when it is fired
(b) the action of striking a tennis ball with a racquet.

NAME: PHYSICS I

SUBJECT: _____

MARK THE LETTER WHICH CORRESPONDS TO THE MOST APPROPRIATE ANSWER
WITH A CROSS

1	A	B	C	D	E
2	A	B	C	D	E
3	A	B	C	D	E
4	A	B	C	D	E
5	A	B	C	D	E
6	A	B	C	D	E
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9	A	B	C	D	E
10	A	B	C	D	E
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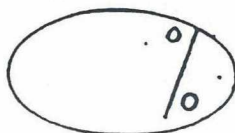
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37	A	B	C	D	E
38	A	B	C	D	E
39	A	B	C	D	E
40	A	B	C	D	E

MARKS	
SECTION A	
SECTION B	
SECTION C	
TOTAL	

PAPER NUMBER



PERCENTAGE



10 SCIENCE
PHYSICS 1
SECTION B

1. (a) kg, m, s.

(b) FUNDAMENTAL UNIT: describes all other units

DERIVED UNIT: based on the fundamental units.

(c) velocity, acceleration, force, etc. (3)

2. (a) (i) 2.732×10^3 (ii) 1.44×10^{-3} (iii) 6.2542×10^3 (iv) 7.462×10^7 (2)

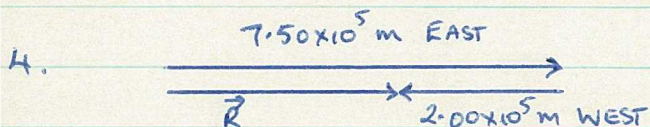
(b) 8.0×10^6 J (1)

(c) 7200 mm, 0.001 km, 76 cm, 0.85 cm. (1)

3. (b) Vector, since it is a force that acts downwards at all times. (1)

$$F_w = mg$$
$$\Rightarrow m = \frac{7.70 \times 10^2}{9.80}$$
$$= 78.57 \text{ kg}$$

$$\therefore \underline{m = 78.6 \text{ kg}} \quad (2)$$



$$\vec{R} = 5.50 \times 10^5 \text{ m EAST.}$$

$$(a) \text{ DISTANCE} = 7.50 \times 10^5 + 2.00 \times 10^5$$
$$= \underline{9.50 \times 10^5 \text{ m}} \quad (1)$$

$$(b) \text{ speed} = \frac{d}{t}$$
$$= \frac{9.50 \times 10^5}{1.44 \times 10^4}$$
$$= 65.97 \text{ ms}^{-1}$$

$$\therefore \underline{\text{speed} = 66.0 \text{ ms}^{-1}} \quad (1)$$

$$(c) \underline{\vec{R} = 5.50 \times 10^5 \text{ m EAST.}} \quad (2)$$

$$(d) v = \frac{s}{t}$$
$$= \frac{5.50 \times 10^5}{1.44 \times 10^4}$$
$$= 38.19 \text{ ms}^{-1}$$

$$\therefore \underline{\text{average velocity} = 38.2 \text{ ms}^{-1} \text{ EAST.}} \quad (2)$$

5. $v = 5.00 \text{ ms}^{-1}$ FORWARDS +ve
 $u = 25.0 \text{ ms}^{-1}$ $a = \frac{v-u}{t}$
 $a = ?$ $= \frac{5.00 - 25.0}{8.00}$
 $t = 8.00 \text{ s}$ $= -2.50 \text{ ms}^{-2}$
 $s = ?$ $\therefore \text{acceleration} = 2.50 \text{ ms}^{-2} \text{ backwards.} \quad (3)$

6. $v = ?$ $\downarrow +ve.$
 $u = 0.00 \text{ ms}^{-1}$ (a) $v = u + at$
 $a = 10.0 \text{ ms}^{-2}$ $= 0 + (10.0)(3.60)$
 $t = 3.60 \text{ s}$ $= 36.0 \text{ ms}^{-1}$
 $s = ?$ $\therefore \text{velocity} = 36.0 \text{ ms}^{-1} \text{ downwards.} \quad (2)$

(b) $s = ut + \frac{1}{2}at^2$
 $= 0 + \frac{1}{2}(10.0)(3.60)^2$
 $= 64.8 \text{ m}$
 $\therefore \text{height} = 64.8 \text{ m.} \quad (2)$

7. (a). An object will remain in a state of rest or uniform motion unless acted upon by an external unbalanced force, and will accelerate in the direction of that force. (1)

(b) $F = ma. \quad (1)$

(c) $F = ma$
 $= (0.400)(20.0)$
 $= 8.00 \text{ N}$

$\therefore \text{force} = 8.00 \text{ N in the direction of the ball's movement.} \quad (2)$

(d)(i) ACTION: bullet pushed forwards REACTION: gun moves backwards
(ii) ACTION: strings are pushed backwards by the ball.
REACTION: strings push the ball forwards. (2)

